

THIEF RIVER FALLS, MN, USA

WMO: 727555

Lat: 48.067N

Lon: 96.183W

Elev: 340

StdP: 97.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94956

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.1	(c) -27.5	(d) -33.2	(e) 0.2	(f) -26.8	(g) -32.2	(h) 0.2	(i) -25.6	(j) 13.1	(k) -12.2	(l) 12.0	(m) -12.8	(n) 2.7	(o) 300	(p) 0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 29.9	(d) 21.4	(e) 28.0	(f) 20.3	(g) 27.1	(h) 19.6	(i) 23.1	(j) 28.0	(k) 21.8	(l) 26.8	(m) 20.6	(n) 25.4	(o) 6.2	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.7	26.5	20.1	15.4	25.2	18.8	14.2	23.9	70.3	28.1	65.3	26.5	61.0	25.7	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.7	(b) 10.5	(c) 9.1	(e) -33.0	(f) 32.9	(g) 3.5	(h) 1.6	(i) -35.5	(j) 34.1	(k) -37.6	(l) 35.0	(m) -39.5	(n) 35.9	(o) -42.0	(p) 37.1		
(4) 11.7	(5) 10.5	(6) 9.1	(7) -31.4	(8) 25.3	(9) 2.1	(10) 1.1	(11) -33.0	(12) 26.2	(13) -34.2	(14) 26.8	(15) -35.4	(16) 27.5	(17) -36.9	(18) 28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-13.9	-11.8	-4.2	4.6	12.0	17.6	20.0	18.9	14.2	6.6	-2.5	-10.6	(6)
(7)		DBStd	13.35	8.07	7.67	7.47	5.75	4.90	3.60	3.05	3.21	4.71	5.10	6.41	7.36	(7)
(8)		HDD10.0	3162	742	611	440	176	35	1	0	0	14	131	375	638	(8)
(9)		HDD18.3	5286	1001	844	697	412	202	54	19	32	140	364	624	896	(9)
(10)		CDD10.0	1093	0	0	1	14	98	229	309	276	139	26	1	0	(10)
(11)		CDD18.3	174	0	0	0	0	7	33	69	49	15	1	0	0	(11)
(12)		CDH23.3	1547	0	0	0	4	91	290	570	429	150	13	0	0	(12)
(13)		CDH26.7	358	0	0	0	1	21	66	134	100	34	2	0	0	(13)
(14)	Wind	WSAvg	4.5	4.6	4.6	4.7	4.9	4.9	4.2	3.7	3.8	4.3	4.9	4.8	4.7	(14)
(15)	Precipitation	PrecAvg	552	12	10	18	31	69	96	80	67	57	44	22	17	(15)
(16)		PrecMax	718	27	32	57	68	164	208	142	167	143	109	81	38	(16)
(17)		PrecMin	305	1	0	1	1	20	24	30	25	2	4	2	0	(17)
(18)		PrecStd	97	9	8	14	15	35	42	31	34	34	28	20	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	6.1	17.1	22.8	29.0	31.4	32.1	32.3	30.2	24.9	15.0	4.1	(19)
(20)			MCWB	1.2	4.0	11.9	13.0	18.3	22.1	22.9	23.1	21.3	16.0	9.9	1.9	(20)
(21)		2%	DB	1.1	2.9	11.3	19.0	26.1	28.0	29.2	29.0	27.3	21.0	11.2	2.4	(21)
(22)			MCWB	0.0	1.0	7.9	10.9	15.6	20.0	22.1	20.9	18.7	14.2	7.7	0.8	(22)
(23)		5%	DB	-1.1	1.4	7.6	17.0	22.9	26.6	27.8	27.4	24.9	17.4	8.0	1.0	(23)
(24)			MCWB	-1.8	0.0	4.8	9.5	14.2	18.7	21.1	19.9	17.5	12.0	5.3	-0.2	(24)
(25)		10%	DB	-2.7	-0.9	4.9	13.8	21.0	24.8	27.0	26.0	22.5	14.3	6.0	-1.1	(25)
(26)			MCWB	-3.7	-1.9	2.5	8.0	13.2	17.8	20.4	18.8	16.3	9.9	3.4	-2.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	4.0	12.3	14.5	19.8	23.7	25.1	24.0	22.2	17.1	10.9	2.8	(27)
(28)			MCD B	3.0	5.7	14.9	20.7	26.0	28.5	29.6	29.4	28.1	22.5	12.9	3.9	(28)
(29)		2%	WB	-0.1	1.4	7.9	11.6	17.6	21.7	23.6	22.3	20.2	14.9	7.9	0.9	(29)
(30)			MCD B	0.9	2.9	10.9	17.7	23.1	26.8	28.1	27.8	25.3	19.9	10.1	2.1	(30)
(31)		5%	WB	-1.9	0.1	4.9	10.0	15.7	20.0	22.3	20.9	18.5	12.5	5.5	-0.3	(31)
(32)			MCD B	-1.3	1.4	7.5	15.6	21.1	24.9	26.8	25.9	23.2	16.8	8.1	0.7	(32)
(33)		10%	WB	-3.6	-1.8	2.8	8.4	14.1	18.8	21.1	19.8	17.1	10.3	3.5	-2.0	(33)
(34)			MCD B	-2.6	-0.9	4.7	13.5	19.2	23.4	25.5	24.3	21.5	14.1	5.6	-1.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	9.2	9.1	10.2	11.1	10.2	10.7	11.3	11.0	9.1	7.8	7.8	(35)
(36)			MCD BR	8.6	8.4	11.0	14.9	14.7	12.7	12.3	13.0	14.3	13.5	10.7	8.3	(36)
(37)			MCWBR	7.8	7.2	8.1	8.7	7.6	6.9	7.0	6.8	7.7	8.2	7.8	7.2	(37)
(38)		5% WB	MCD BR	8.3	7.9	10.8	13.6	12.4	10.8	11.2	11.8	12.1	12.3	10.2	8.3	(38)
(39)			MCWBR	7.6	6.8	8.2	8.5	7.5	6.9	7.3	6.8	7.4	7.9	7.7	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.276	0.302	0.294	0.344	0.361	0.376	0.378	0.379	0.348	0.303	0.276	0.266	(40)	
(41)		taud	2.279	2.209	2.426	2.371	2.383	2.368	2.370	2.375	2.454	2.548	2.510	2.383	(41)	
(42)		Ebn at Noon	793	847	931	907	901	886	878	862	859	855	811	771	(42)	
(43)		Edh at Noon	74	100	95	112	115	118	116	111	93	72	59	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.54	2.75	3.90	4.85	5.45	5.89	6.28	5.24	3.85	2.36	1.38	1.07	(44)	
(45)		RadStd	0.16	0.23	0.40	0.44	0.34	0.44	0.28	0.37	0.28	0.34	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	(47) Regional (25 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(46) Station Only	(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page